Macroeconomics #7

Dynamic General Equilibrium and Real Business Cycles

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Today's lecture

1. Intro
2. Model
3. Steady state
4. Log-linearization
5. Some analyses

What is the RBC Model?

The Real Business Cycle (RBC) model is the neoclassical growth model augmented with:

- **Variable labor supply** — households choose how much to work
- **Stochastic productivity shocks** — the driving force of business cycles

Key references:

- Kydland and Prescott (1982) — "Time to Build and Aggregate Fluctuations"
- King, Plosser, and Rebelo (1988)
- King and Rebelo (2000)

Central question: Can a simple, frictionless model driven only by technology shocks replicate the key features of business cycle data?

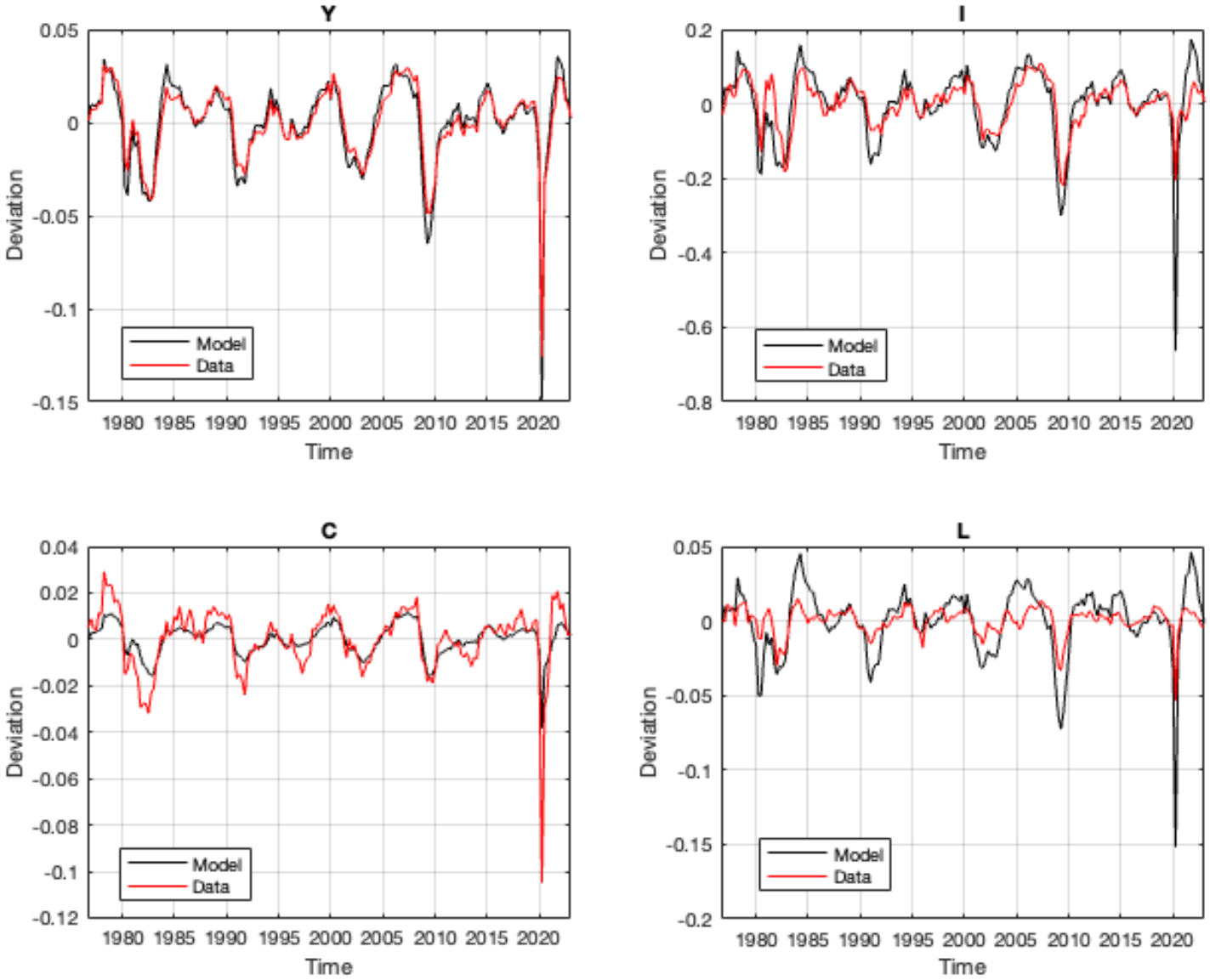


Figure: RBC versus Postwar US data

Agenda

Building and analyzing the RBC model involves several steps:

1. **Build the model:** Specify the model's components, derive optimal solutions for each party
2. **Derive key equations:** general equilibrium
3. **Solve the model:** Log-linearization, then solve the state-space representation.
4. **Calibrate/Estimate:** assign parameter values using data targets, find steady states
5. **Evaluate:** compare model moments and impulse responses with data

Model

Environment

- Closed economy with a representative household and a representative firm
- Firm uses labor N_t and capital K_t to produce output Y_t
- Household supplies labor and capital, chooses consumption and leisure
- All values are real (no nominal frictions)

Two equivalent formulations:

1. Firms own capital; households hold bonds and receive dividends
2. Households own capital and rent it to firms

Both yield identical equilibrium conditions (Modigliani–Miller).

Production Technology

The representative firm produces output using a CRS production function:

$$Y_t = A_t F(K_t, N_t)$$

where A_t is total factor productivity (TFP).

The production function satisfies:

- Essentiality: $F(0, N) = F(K, 0) = 0$
- Strict monotonicity and concavity:
  $F_K > 0, F_{KK} < 0, F_N > 0, F_{NN} < 0$
- Inada conditions: $\lim_{K \to 0^+} F_K = +\infty$, $\lim_{K \to \infty} F_K = 0$

TFP follows an AR(1) process in logs:

$$\ln A_t = \rho \ln A_{t-1} + s_A \varepsilon_t, \quad \varepsilon_t \sim N(0, 1), \quad |\rho| < 1$$

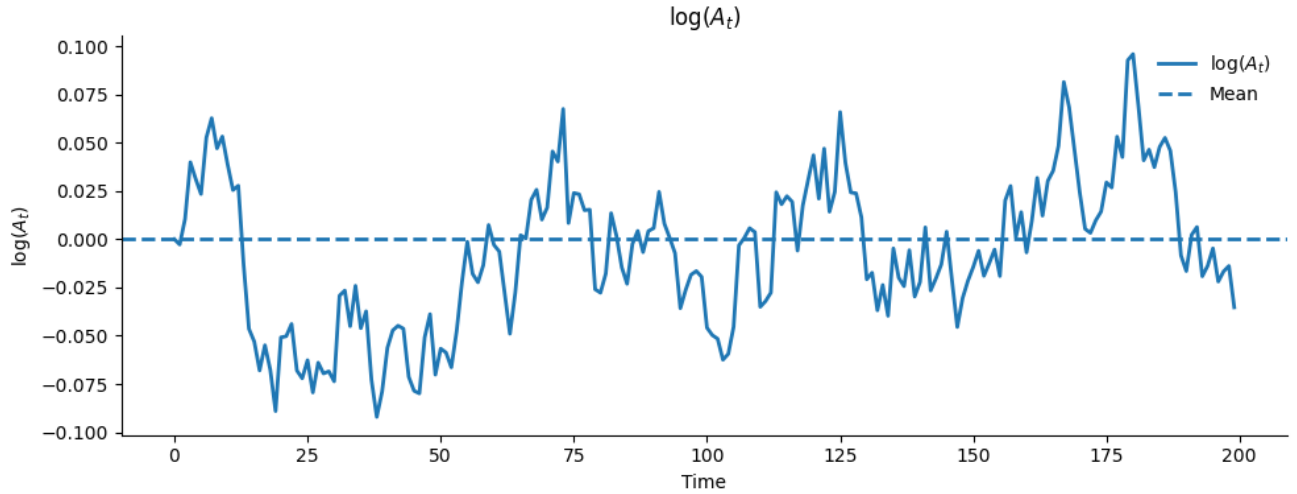


Figure: log form

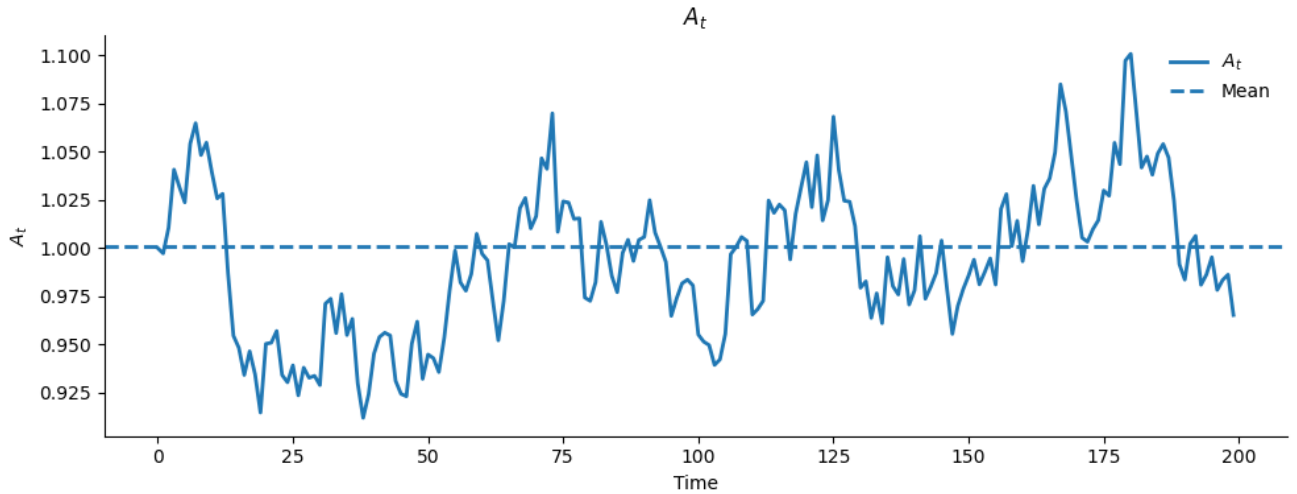


Figure: level

Firm's Profit Maximization

The firm maximizes profit $\Pi_t = A_t F(K_t, N_t) - w_t N_t - R_t^k K_t$.

First-order conditions (factor demands):

$$w_t = A_t F_N(K_t, N_t) \tag{1}$$

$$R_t^k = A_t F_K(K_t, N_t) \tag{2}$$

With Cobb–Douglas production $F(K, N) = K^\alpha N^{1-\alpha}$:

$$w_t = (1-\alpha) A_t K_t^\alpha N_t^{-\alpha}$$

$$R_t^k = \alpha A_t K_t^{\alpha-1} N_t^{1-\alpha}$$

Household

A representative household maximizes expected lifetime utility:

$$\max_{C_t, N_t, K_{t+1}} \quad \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \, u(C_t, 1 - N_t)$$

where $\beta \in (0, 1)$ is the discount factor, C_t is consumption, N_t is labor, and $1 - N_t$ is leisure (time endowment normalized to 1).

The budget constraint (households own capital):

$$C_t + K_{t+1} - (1 - \delta)K_t = w_t N_t + R_t^k K_t$$

or equivalently:

$$C_t + I_t = w_t N_t + R_t^k K_t$$

with the law of motion for capital:

$$K_{t+1} = I_t + (1 - \delta)K_t$$

Household's First-Order Conditions

Lagrangian

$$\mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Big\{ u(C_t, 1 - N_t) + \lambda_t [w_t N_t + R_t^k K_t + (1-\delta) K_t - C_t - K_{t+1}] \Big\}$$

FOC

$$(C_t): \qquad \frac{\partial \mathcal{L}}{\partial C_t} = u_C - \lambda_t = 0,$$

$$(N_t): \qquad \frac{\partial \mathcal{L}}{\partial N_t} = -u_l + \lambda_t w_t = 0,$$

$$(K_{t+1}): \qquad \frac{\partial \mathcal{L}}{\partial K_{t+1}} = -\lambda_t + \beta\, \mathbb{E}_t \{ \lambda_{t+1} [R_{t+1}^k + (1-\delta)] \} = 0,$$

The FOC for consumption, at t and $t+1$, gives

$$u_C(C_t, 1 - N_t) = \lambda_t,$$

$$u_C(C_{t+1}, 1 - N_{t+1}) = \lambda_{t+1}$$

Consumption–leisure trade-off:

$$\frac{u_l(C_t, 1 - N_t)}{u_C(C_t, 1 - N_t)} = w_t \tag{3}$$

The marginal rate of substitution between leisure and consumption equals the real wage.

Euler equation for capital:

$$u_C(C_t, 1 - N_t) = \beta \, \mathbb{E}_t \left[u_C(C_{t+1}, 1 - N_{t+1}) \left(R^k_{t+1} + 1 - \delta \right) \right] \tag{4}$$

The marginal utility of consuming today equals the expected discounted marginal utility from saving one more unit, earning the return $R^k_{t+1} + 1 - \delta$, and consuming tomorrow.

Competitive Equilibrium

Definition. Given K_0 and $\{A_t\}_{t=0}^{\infty}$, a competitive equilibrium is a set of prices $\{w_t, R_t^k\}$ and allocations $\{C_t, N_t, K_{t+1}, Y_t, I_t\}$ such that:

1. Households optimize (Euler equation + labor supply)
2. Firms optimize (factor demand conditions)
3. Goods market clears: $Y_t = C_t + I_t$

The **equilibrium system** (8 equations, 8 unknowns):

1. Euler equation
2. Labor supply
3. Capital accumulation
4. TFP process
5. Production function
6. Resource constraint
7. Wage equation
8. Capital rental rate

Functional Form

Preferences:

$$u(C_t, N_t) = \frac{C_t^{1-\eta}}{1-\eta} - \theta \frac{N_t^{1+\varphi}}{1+\varphi}$$

- $\eta > 0$: inverse of the intertemporal elasticity of substitution (IES $= 1/\eta$)
- $\varphi \geq 0$: inverse of the Frisch labor supply elasticity
- $\theta > 0$: weight on labor disutility
- When $\eta = 1$: $u(C) = \ln C$ (log utility)

FOC

$$\frac{u_l}{u_C} = \frac{\theta N_t^{\varphi}}{C_t^{-\eta}} = \theta C_t^{\eta} N_t^{\varphi} = w_t$$

Consumption

- High η, low IES, consumption is less responsive to change in interest, households prefer consumption smoothing
- Low η, high IES, consumption is more responsive to change in interest, households are more flexible in changing consumption

Labor

- High φ, Frisch elasticity $1/\varphi$ is low, labor is less elastic, workers dislike additional work, so they are less responsive to wage changes
- Low φ, Frisch elasticity $1/\varphi$ is high, labor is more elastic, workers adjust hours easily, so they are more responsive to wage changes

Production:

$$Y_t = A_t K_t^{\alpha} N_t^{1-\alpha}, \quad \alpha \in (0,1)$$

Factor prices

$$w_t = (1-\alpha) A_t K_t^{\alpha} N_t^{-\alpha}$$
$$R_t^k = \alpha A_t K_t^{\alpha-1} N_t^{1-\alpha}$$

Goods market clear

$$Y_t = C_t + I_t$$

Equilibrium Conditions with Functional Forms

Substituting the functional forms, the equilibrium system becomes:

$$\begin{aligned}
\text{Euler:} \quad & C_t^{-\eta} = \beta \mathbb{E}_t \left[C_{t+1}^{-\eta} (R_{t+1}^k + 1 - \delta) \right] \\
\text{Labor supply:} \quad & \theta C_t^{\eta} N_t^{\varphi} = w_t \\
\text{Capital accumulation:} \quad & K_{t+1} = (1-\delta) K_t + I_t \\
\text{Production:} \quad & Y_t = A_t K_t^{\alpha} N_t^{1-\alpha} \\
\text{Capital demand:} \quad & R_t^k = \alpha A_t K_t^{\alpha-1} N_t^{1-\alpha} \\
\text{Labor demand:} \quad & w_t = (1-\alpha) A_t K_t^{\alpha} N_t^{-\alpha} \\
\text{Resource constraint:} \quad & Y_t = C_t + I_t \\
\text{TFP process:} \quad & \ln A_t = \rho \ln A_{t-1} + s_A \varepsilon_t
\end{aligned}$$

Steady State

Non-Stochastic Steady State (I)

In the steady state: $K_{t+1} = K_t = K$, $C_{t+1} = C_t = C$, $A = 1$, $\varepsilon = 0$.

Step 1: From the Euler equation (uncertainty vanishes):

$$\overline{R} = \frac{1}{\beta} - 1 + \delta$$

The steady-state rental rate is pinned down by preferences and depreciation alone.

Step 2: From capital demand $\overline{R} = \alpha(K/N)^{\alpha-1}$:

$$\frac{K}{N} = \left(\frac{\alpha}{\overline{R}}\right)^{\frac{1}{1-\alpha}}$$

Step 3: Steady-state wage:

$$w = (1-\alpha)\left(\frac{K}{N}\right)^{\alpha}$$

Non-Stochastic Steady State (II)

Step 4: From capital accumulation, $I = \delta K$, so:

$$\frac{I}{N} = \delta \frac{K}{N}$$

Step 5: From the resource constraint:

$$\frac{C}{N} = \left(\frac{K}{N}\right)^{\alpha} - \delta \frac{K}{N}$$

Step 6: Set target $N = 1/3$ (empirical target), then back out θ:

$$\theta = \frac{w}{C^{\eta} N^{\varphi}}$$

With N known, all steady-state values $(K, Y, C, I, w, \overline{R})$ follow suit.

Key ratios (with $\alpha = 1/3$, $\beta = 0.99$, $\delta = 0.02$):
$I/Y = \alpha\delta/(1/\beta - 1 + \delta) \approx 0.22, \quad C/Y \approx 0.78.$

Log-linearization

Why Log-Linearize?

The equilibrium system is a set of **nonlinear stochastic difference equations**. No closed-form solution exists in general.

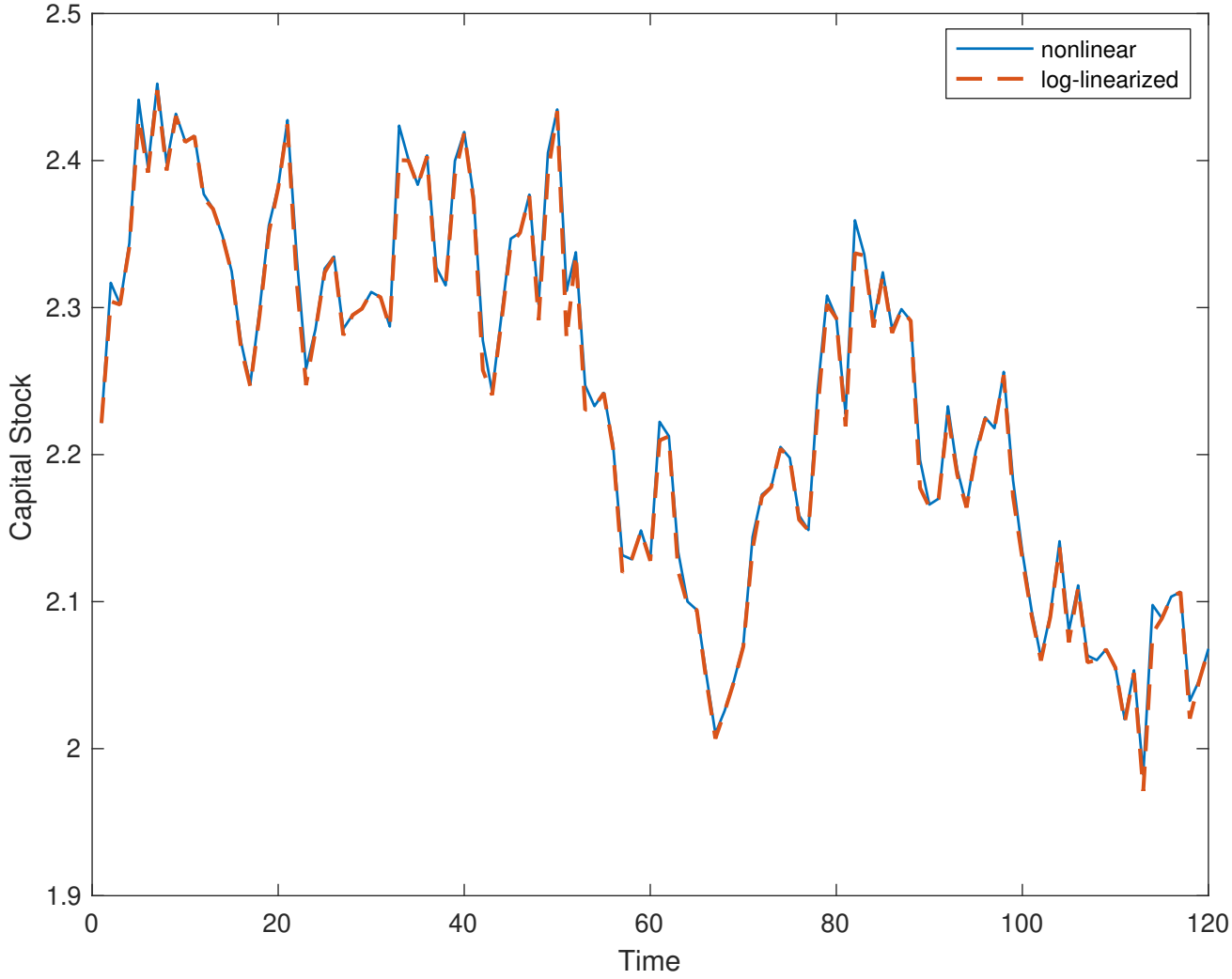
Strategy: Approximate each equation around the non-stochastic steady state using a first-order Taylor expansion in log-deviations.

Define the log-deviation of variable X_t:

$$\widetilde{X}_t = \ln X_t - \ln \overline{X} \approx \frac{X_t - \overline{X}}{\overline{X}}$$

Uhlig's method: substitute $X_t = \overline{X}\, e^{\widetilde{X}_t}$ and use the approximation rules:

$$\begin{aligned} e^{\widetilde{X}_t} &\approx 1 + \widetilde{X}_t \\ e^{\widetilde{X}_t + a\widetilde{Y}_t} &\approx 1 + \widetilde{X}_t + a\widetilde{Y}_t \\ \widetilde{X}_t \cdot \widetilde{Y}_t &\approx 0 \end{aligned}$$


nonlinear
log-linearized
Capital Stock
Time
2.5
2.4
2.3
2.2
2.1
2
1.9
0
20
40
60
80
100
120

$$\text{Euler:} \quad C_t^{-\eta} = \beta \mathbb{E}_t \left[C_{t+1}^{-\eta} (R_{t+1}^k + 1 - \delta) \right]$$

$$\frac{\eta}{\beta} \mathbb{E}_t (\widetilde{C}_{t+1} - \widetilde{C}_t) = \overline{R} \, \mathbb{E}_t \widetilde{R}_{t+1}^k$$

Labor supply: $\theta C_t^{\eta} N_t^{\varphi} = w_t$

$$\eta \widetilde{C}_t + \varphi \widetilde{N}_t = \widetilde{w}_t$$

Capital accumulation: $K_{t+1} = (1-\delta)K_t + I_t$

$$\widetilde{K}_{t+1} = (1-\delta)\widetilde{K}_t + \delta\,\widetilde{I}_t$$

Production: $Y_t = A_t K_t^{\alpha} N_t^{1-\alpha}$

$$\widetilde{Y}_t = \widetilde{A}_t + \alpha \widetilde{K}_t + (1-\alpha)\widetilde{N}_t$$

Capital demand: $R_t^k = \alpha A_t K_t^{\alpha-1} N_t^{1-\alpha}$

$$\widetilde{R}_t^k = \widetilde{Y}_t - \widetilde{K}_t$$

Labor demand: $w_t = (1-\alpha) A_t K_t^{\alpha} N_t^{-\alpha}$

$$\widetilde{w}_t = \widetilde{Y}_t - \widetilde{N}_t$$

Resource constraint: $Y_t = C_t + I_t$

$$\overline{Y}\,\widetilde{Y}_t = \overline{C}\,\widetilde{C}_t + \overline{I}\,\widetilde{I}_t$$

TFP process: $\ln A_t = \rho \ln A_{t-1} + s_A \varepsilon_t$

$$\widetilde{A}_t = \rho \widetilde{A}_{t-1} + s_A \varepsilon_t$$

Linearized Equilibrium System

We get a system of **8 linear equations** in 8 log-deviation variables: $\widetilde{Y}_t, \widetilde{C}_t, \widetilde{N}_t, \widetilde{w}_t, \widetilde{R}^k_t, \widetilde{I}_t, \widetilde{K}_t, \widetilde{A}_t$. Their steady states are zero.

$$\text{Euler:} \quad \frac{\eta}{\beta}\,\mathbb{E}_t(\widetilde{C}_{t+1} - \widetilde{C}_t) = \overline{R}\,\mathbb{E}_t\widetilde{R}^k_{t+1}$$

$$\text{Labor supply:} \quad \eta\widetilde{C}_t + \varphi\widetilde{N}_t = \widetilde{w}_t$$

$$\text{Capital accumulation:} \quad \widetilde{K}_{t+1} = (1-\delta)\widetilde{K}_t + \delta\,\widetilde{I}_t$$

$$\text{Production:} \quad \widetilde{Y}_t = \widetilde{A}_t + \alpha\widetilde{K}_t + (1-\alpha)\widetilde{N}_t$$

$$\text{Capital demand:} \quad \widetilde{R}^k_t = \widetilde{Y}_t - \widetilde{K}_t$$

$$\text{Labor demand:} \quad \widetilde{w}_t = \widetilde{Y}_t - \widetilde{N}_t$$

$$\text{Resource constraint:} \quad \overline{Y}\,\widetilde{Y}_t = \overline{C}\,\widetilde{C}_t + \overline{I}\,\widetilde{I}_t$$

$$\text{TFP process:} \quad \widetilde{A}_t = \rho\,\widetilde{A}_{t-1} + s_A\varepsilon_t$$

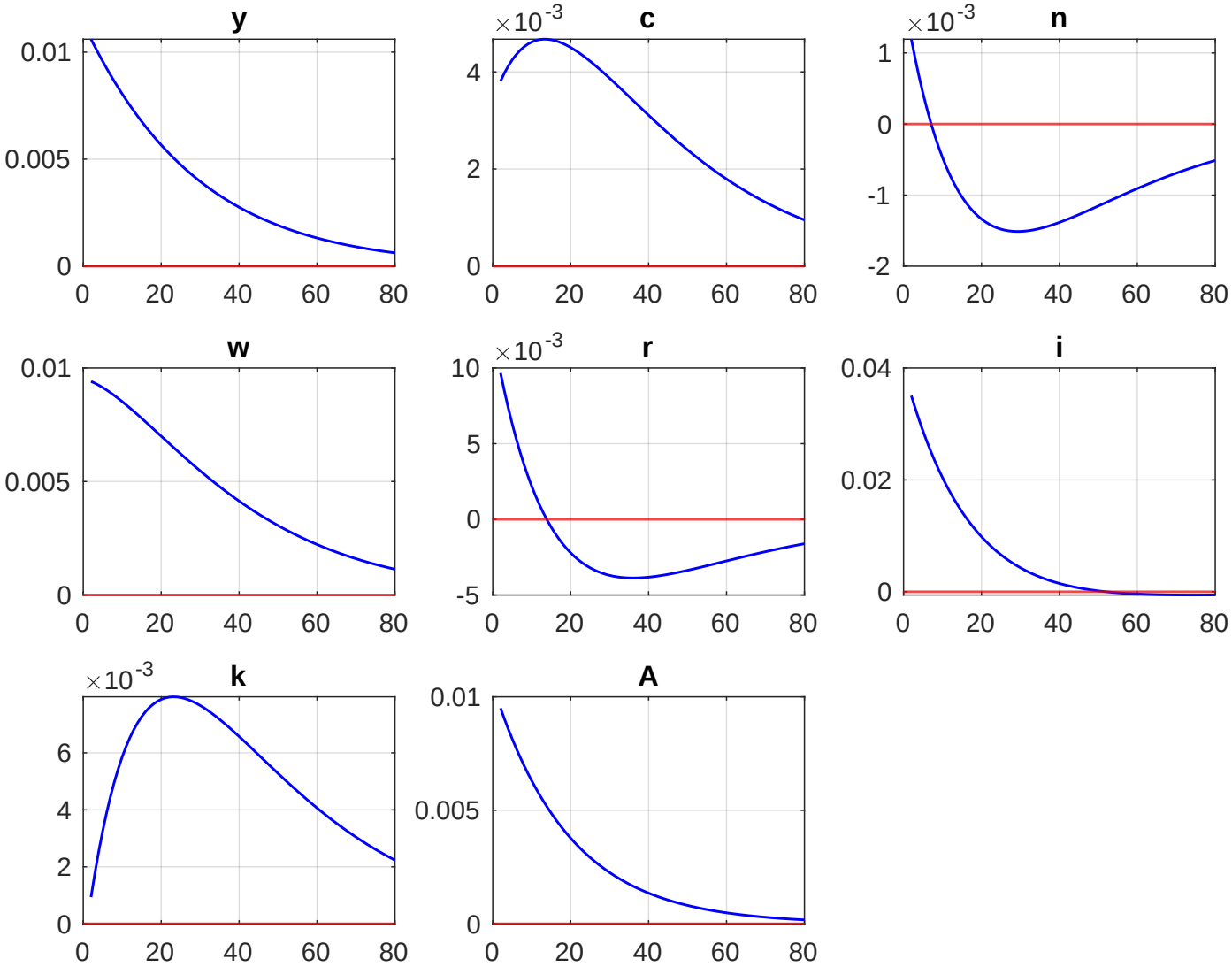
Temporary Productivity Shock

A purely temporary (iid) shock to $\widetilde{A}_t$: lasts only one period.

Key differences from permanent shock:

- Consumption barely reacts (permanent income logic)
- Labor rises more (smaller income effect $\Rightarrow$ substitution effect dominates)
- Output rises more in relative terms (stronger labor amplification)
- Real wage rises less
- R_t^k rises on impact, quickly returns
- All variables return to steady state quickly

General principle: More persistent shocks $\Rightarrow$ responses look more like permanent case. More transitory $\Rightarrow$ responses look more like iid case. Standard calibration ($\rho \approx 0.98$) falls between the two.


y
0.01
0.005
0
0 20 40 60 80
c
×10^-3
4
2
0
0 20 40 60 80
n
×10^-3
1
0
-1
-2
0 20 40 60 80
w
0.01
0.005
0
0 20 40 60 80
r
×10^-3
10
5
0
-5
0 20 40 60 80
i
0.04
0.02
0
0 20 40 60 80
k
×10^-3
6
4
2
0
0 20 40 60 80
A
0.01
0.005
0
0 20 40 60 80

Calibration Strategy

Idea: Choose parameters so the model's **steady state** matches long-run features of the data. Then use the *same* model to explain short-run fluctuations.

Parameter	Target / Source	Value
α	Capital's share of income	$1/3$
β	Average real interest rate $\approx 4\%$/yr	0.99 (quarterly)
δ	Investment–output ratio $\approx 22\%$	0.02 (quarterly)
θ	Steady-state $N = 1/3$ ($\eta = \varphi = 1$)	≈ 7.71
ρ	AR(1) of linearly detrended TFP	0.979
s_A	Std. dev. of TFP innovations	0.009

Stylized US Business Cycle Facts

HP-filtered US data (1947Q1–2022Q3), $\lambda = 1600$:

Series	Std. Dev.	Rel. Std. Dev.	Corr w/ Y_t
Output	0.0166	1.00	1.00
Consumption	0.0110	0.66	0.76
Investment	0.0444	2.67	0.76
Hours	0.0211	1.27	0.87
Avg. Labor Productivity	0.0114	0.69	0.27
Wage	0.0115	0.69	−0.01
Real Rate	0.0043	0.26	−0.00
TFP	0.0127	0.76	0.78

Key facts: consumption is smoother than output, investment is $\sim$2.5$\times$ more volatile, hours are procyclical, all series are persistent.

Model vs. Data

	Rel. Std. Dev.		correlation with Y_t	
Series	Data	Model	Data	Model
Consumption	0.66	0.38	0.76	0.96
Investment	2.67	3.25	0.76	0.99
Hours	1.27	0.43	0.87	0.98
Avg. Labor Prod.	0.69	0.58	0.27	0.99
Wage	0.69	0.58	−0.01	0.99
Real Rate	0.26	0.03	−0.00	0.97
TFP	0.76	0.71	0.78	1.00

Successes: output volatility, relative volatility of consumption and investment, persistence, procyclicality of main aggregates.

Failures: hours too smooth, correlations too high, wage and labor productivity perfectly correlated (data: nearly zero).

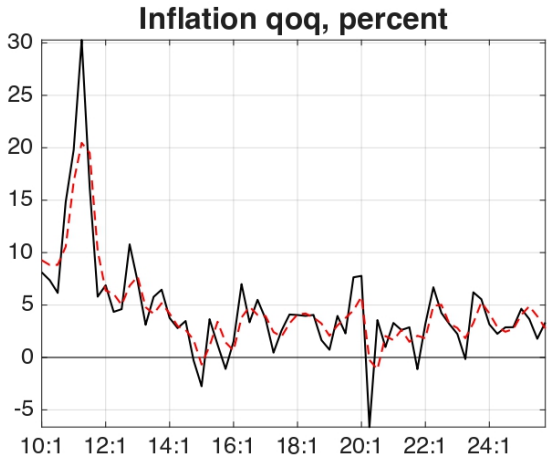
Where Do We Go From Here?

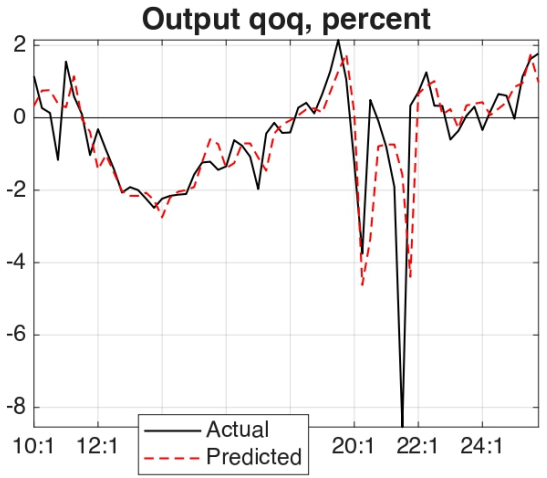
The RBC model is the **backbone** of modern DSGE models.
Extensions include:

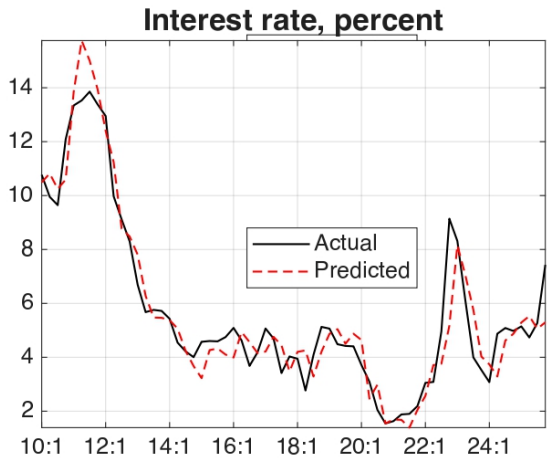
- **New Keynesian models:** add sticky prices/wages ⇒ monetary policy matters
- **Multiple shocks:** government spending, investment-specific technology, preference shocks
- **Labor market frictions:** search and matching (Mortensen–Pissarides)
- **Financial frictions:** Bernanke–Gertler–Gilchrist financial accelerator
- **Variable capital utilization:** helps with amplification
- **Habit formation:** generates consumption persistence

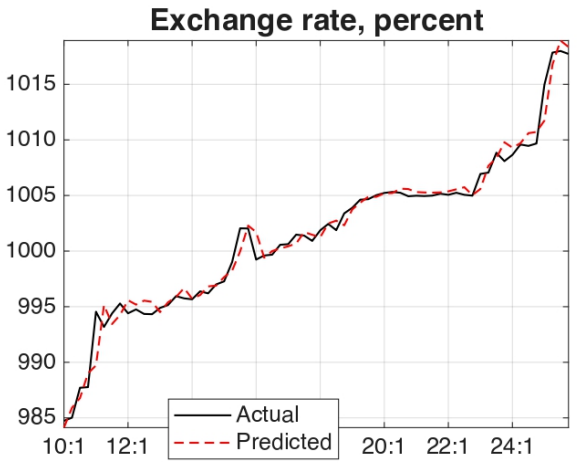
The methodology (DSGE modeling, log-linearization, calibration, impulse responses) remains the standard approach in quantitative macroeconomics.

Viet Nam


Inflation qoq, percent
30
25
20
15
10
5
0
-5
10:1
12:1
14:1
16:1
18:1
20:1
22:1
24:1


Output qoq, percent
2
0
-2
-4
-6
-8
10:1
12:1
20:1
22:1
24:1
Actual
Predicted


Interest rate, percent
14
12
10
8
6
4
2
10:1
12:1
14:1
16:1
18:1
20:1
22:1
24:1
Actual
Predicted


Exchange rate, percent
1015
1010
1005
1000
995
990
985
10:1
12:1
20:1
22:1
24:1
Actual
Predicted

Appendix

Appendix: Reducing the Linear System

Hours. Labour supply $\widetilde{w}_t = \eta \widetilde{C}_t + \varphi \widetilde{N}_t$ and labour demand $\widetilde{w}_t = \widetilde{Y}_t - \widetilde{N}_t$ give

$$\widetilde{N}_t = \frac{\widetilde{Y}_t - \eta \widetilde{C}_t}{1+\varphi}.$$

Output. Substitute into $\widetilde{Y}_t = \widetilde{A}_t + \alpha \widetilde{K}_t + (1-\alpha)\widetilde{N}_t$ and use $1 - \frac{1-\alpha}{1+\varphi} = \frac{\alpha+\varphi}{1+\varphi}$:

$$\widetilde{Y}_t = \psi_A \widetilde{A}_t + \psi_K \widetilde{K}_t - \psi_C \widetilde{C}_t, \quad \psi_A = \frac{1+\varphi}{\alpha+\varphi},\ \psi_K = \frac{\alpha(1+\varphi)}{\alpha+\varphi},\ \psi_C = \frac{(1-\alpha)\eta}{\alpha+\varphi}.$$

Capital. Use $\widetilde{I}_t = (\overline{Y}\widetilde{Y}_t - \overline{C}\widetilde{C}_t)/\overline{I}$ and $\delta/\overline{I} = 1/\overline{K}$ in $\widetilde{K}_{t+1} = (1-\delta)\widetilde{K}_t + \delta \widetilde{I}_t$:

$$\widetilde{K}_{t+1} = \gamma_K \widetilde{K}_t + \gamma_A \widetilde{A}_t - \gamma_C \widetilde{C}_t, \quad \gamma_K = 1-\delta+\frac{\overline{Y}}{\overline{K}}\psi_K,\ \gamma_A = \frac{\overline{Y}}{\overline{K}}\psi_A,\ \gamma_C = \frac{\overline{Y}}{\overline{K}}\psi_C + \frac{\overline{C}}{\overline{K}}.$$

Euler. With $\widetilde{R}^k_{t+1} = \widetilde{Y}_{t+1} - \widetilde{K}_{t+1}$ and $\mathbb{E}_t \widetilde{A}_{t+1} = \rho \widetilde{A}_t$:

$$\eta\, \mathbb{E}_t(\widetilde{C}_{t+1} - \widetilde{C}_t) = \beta \overline{R}\, \mathbb{E}_t\left[\psi_A \widetilde{A}_{t+1} + (\psi_K - 1)\widetilde{K}_{t+1} - \psi_C \widetilde{C}_{t+1}\right].$$

Appendix: Undetermined Coefficients

Guess a linear solution in the states $(\widetilde{K}_t, \widetilde{A}_t)$:

$$\widetilde{C}_t = \nu_{CK}\widetilde{K}_t + \nu_{CA}\widetilde{A}_t, \qquad \widetilde{K}_{t+1} = \nu_{KK}\widetilde{K}_t + \nu_{KA}\widetilde{A}_t.$$

Capital equation: substituting the consumption guess,

$$\nu_{KK} = \gamma_K - \gamma_C\nu_{CK}, \qquad \nu_{KA} = \gamma_A - \gamma_C\nu_{CA}.$$

Euler equation: substitute both guesses; it must hold for *all* $\widetilde{K}_t$ and $\widetilde{A}_t$, so the coefficients on each must match.

- Coefficient on $\widetilde{K}_t$ — a quadratic in ν_{CK}:

$$\eta\,\nu_{CK}(\nu_{KK} - 1) = \beta\overline{R}\big[(\psi_K - 1) - \psi_C\nu_{CK}\big]\nu_{KK}.$$

  Keep the root with $|\nu_{KK}| < 1$ (the other root is explosive and violates the TVC).

- Coefficient on $\widetilde{A}_t$ — linear in ν_{CA}, given ν_{CK}:

$$\eta\big[\nu_{CK}\nu_{KA} + (\rho - 1)\nu_{CA}\big] = \beta\overline{R}\Big[\psi_A\rho + (\psi_K - 1)\nu_{KA} - \psi_C\big(\nu_{CK}\nu_{KA} + \rho\nu_{CA}\big)\Big].$$

Appendix: Numerical Solution

Calibration: $\alpha = 1/3$, $\beta = 0.99$, $\delta = 0.02$, $\eta = \varphi = 1$, $\rho = 0.979$ (so $\overline{R} \approx 0.0301$).

- $\psi_A = 1.5$, $\psi_K = \psi_C = 0.5$; $\overline{Y}/\overline{K} \approx 0.0903$, $\overline{C}/\overline{K} \approx 0.0703$.
- $\gamma_K \approx 1.0252$, $\gamma_A \approx 0.1355$, $\gamma_C \approx 0.1155$.
- Quadratic roots: $\nu_{CK} \approx 0.562$ ($\nu_{KK} \approx 0.960$, stable) and $\nu_{CK} \approx -0.232$ ($\nu_{KK} \approx 1.052$, discarded).
- Then $\nu_{CA} \approx 0.480$ and $\nu_{KA} \approx 0.080$.

Policy functions:

$$\widetilde{K}_{t+1} = 0.960\,\widetilde{K}_t + 0.080\,\widetilde{A}_t, \qquad \widetilde{C}_t = 0.562\,\widetilde{K}_t + 0.480\,\widetilde{A}_t,$$
$$\widetilde{Y}_t = 0.219\,\widetilde{K}_t + 1.260\,\widetilde{A}_t, \qquad \widetilde{N}_t = -0.172\,\widetilde{K}_t + 0.390\,\widetilde{A}_t,$$
$$\widetilde{I}_t = -0.988\,\widetilde{K}_t + 4.000\,\widetilde{A}_t, \qquad \widetilde{w}_t = 0.391\,\widetilde{K}_t + 0.870\,\widetilde{A}_t.$$

On impact of a 1% TFP shock ($\widetilde{K}_0 = 0$): output +1.26%, consumption +0.48%, hours +0.39%, investment +4.0%.

Appendix: Constructing the System Matrices

See **The ABC of RBCs** by George McCandless (2008) for more detail.

The coefficient matrices for the undetermined coefficients method:

$$A = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad B = \begin{pmatrix} -(1-\delta) \\ -\alpha \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad D = \begin{pmatrix} 0 \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$C = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & -\delta \\ 1 & 0 & -(1-\alpha) & 0 & 0 & 0 \\ 0 & \eta & \varphi & -1 & 0 & 0 \\ -1 & 0 & 1 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 & 0 \\ -\overline{Y} & \overline{C} & 0 & 0 & 0 & \overline{I} \end{pmatrix}$$

State-Space Representation

Classify variables:

- **Endogenous state** (predetermined): $x_t = [\widetilde{K}_t]$
- **Exogenous state**: $z_t = [\widetilde{A}_t]$
- **Jump variables**: $y_t = [\widetilde{Y}_t, \widetilde{C}_t, \widetilde{N}_t, \widetilde{w}_t, \widetilde{R}^k_t, \widetilde{I}_t]'$

The linearized system can be written as:

$$\text{Static block:} \quad 0 = Ax_t + Bx_{t-1} + Cy_t + Dz_t$$

$$\text{Forward block:} \quad 0 = \mathbb{E}_t[Fx_{t+1} + Gx_t + Hx_{t-1} + \dots$$

$$+Jy_{t+1} + Ky_t + Lz_{t+1} + Mz_t]$$

$$\text{Exogenous:} \quad z_{t+1} = Nz_t + \varepsilon_{t+1}$$

The **solution** takes the form:

$$\boxed{x_t = Px_{t-1} + Qz_t, \qquad y_t = Rx_{t-1} + Sz_t}$$

Method of Undetermined Coefficients

Idea: Guess the form of the solution, substitute into the equilibrium conditions, and match coefficients.

Substituting the guessed policy functions into the system yields:

$$R = -C^{-1}(AP + B)$$
$$S = -C^{-1}(AQ + D)$$

P solves a **quadratic equation**:

$$(F - JC^{-1}A)\,P^2 - (JC^{-1}B - G + KC^{-1}A)\,P - KC^{-1}B + H = 0$$

Select the root satisfying $|P| < 1$ (stability).

Q is solved from a linear system using the Kronecker product / vec operator.

Blanchard–Kahn Method

Alternative approach: Write the system as

$$B \begin{pmatrix} x_{t+1} \\ \mathbb{E}_t y_{t+1} \end{pmatrix} = A \begin{pmatrix} x_t \\ y_t \end{pmatrix} + G\varepsilon_t$$

Steps:

1. Compute $Z = B^{-1}A$, find eigenvalues and eigenvectors
2. Order eigenvalues: stable ($|\lambda| < 1$) first, unstable ($|\lambda| > 1$) last
3. **BK condition:** # unstable eigenvalues = # jump variables
4. Jump variables: $y_t = -\widehat{M}_{22}^{-1}\widehat{M}_{21}\, x_t$ (deterministic part)
5. State transition:
   $x_{t+1} = [\widehat{M}_{11} - \widehat{M}_{12}\widehat{M}_{22}^{-1}\widehat{M}_{21}]^{-1}\overline{\Lambda}_{11}[\widehat{M}_{11} - \widehat{M}_{12}\widehat{M}_{22}^{-1}\widehat{M}_{21}]\, x_t$

Both methods produce **numerically identical** results.